Hongdao Meng

mycrofthd@gmail.com | github.com/Mycroft-s | linkedin.com/in/hongdao-meng-70222b306 | (718) 3063737

EDUCATION

New York University Sep. 2024 - May 2026 (Expected)
Master of Science in Computer Science GPA: 3.8 New York, NY

Beijing University of Technology Sep. 2020 - Jul. 2024
Bachelor of Engineering in Information Security GPA: 4.0 Beijing, CN

SKILLS

Languages: Java, Python, C/C++, Go, SQL, JavaScript, HTML/CSS, Shell, PHP, LaTeX
Frameworks: React, Angular, Vue.js, Django, Flask, Node.js, Spring Boot, PyTorch, TensorFlow, Pandas, Scikit-Learn
Database: MySQL, Redis, MongoDB, PostgreSQL, DynamoDB, Oracle, Firebase, RocketMQ, Elasticsearch, MilvusDB
Tools: Git, Docker, AWS, Azure, CMake, Postman, CI/CD, Jenkins, Nginx, LangChain, FFmpeg, OpenCV, Jira, Figma

PROFESSIONAL & RESEARCH EXPERIENCE

***Machine Learning Engineer Intern* @ C2SMARTER Center, New York** Jan. 2025 - Present
- Engineered an interactive chatbot using **Python**, **Flask**, **React**, **Docker**, **Redis**, **ChromaDB**, and **RAG**; led the finetuning of the **Mistral 7B** model to specialize it for ITS (Intelligent Transportation Systems) project content, enabling semantic search across domain-specific documents and contributing to a USDOT-aligned knowledge base
- Established a distributed system architecture that integrates **LLM-based question answering** with advanced retrieval pipelines, improving user query relevance and **reducing response latency by 23.5%**
- Implemented **CI/CD** workflows on **AWS** (EC2, RDS) and containerized services with **Docker**, ensuring scalability and high availability for 2k+ daily active users

***Machine Learning Engineer and Founder* @ DeepFake Detection Startup, New York** Sep. 2024 - Dec. 2024
- Led 5-member team to develop core modules of deepfake detection web platform using **React** and **TypeScript** for seamless real-time interaction, enabling 1,200+ concurrent users and **reducing client-side rendering latency by 21.3%**
- Fine-tuned **Vision Transformer** and **VGG16** models from **Hugging Face** for image and audio DeepFake detection, achieving **91.2%** and **88.1%** accuracy, respectively. Constructed two custom datasets from the MP4 **FaceForensics** corpus by extracting video frames with **OpenCV** and audio files with **FFmpeg**, and deployed both models on **AWS EC2**
- Built a real-time communication layer using **Django** and **WebSocket** for robust middleware communication, **reducing task completion time by 25.6%** with <180ms P95 latency, and deployed backend services on **Kubernetes (AWS EKS)** with **AWS ELB** load balancing and **HPA policies**, achieving **99.5% availability under 5k RPM**
- Optimized **PostgreSQL** query execution through composite index tuning, reducing average response time by 18% (320ms→262ms)

***Machine Learning Engineer Intern* @ QingTeng, Cloud Platform R&D Department** Feb. 2024 - Aug. 2024
- Led 6-member team to develop **RAG-based chatbot system** using **LangChain** and **Flask-React**, achieving **24.3% accuracy** improvement on MS MARCO dataset **(F1=0.86) with 33.7% faster response latency** through query optimization
- Built hybrid retrieval framework with **MilvusDB** vector database and **BGE-M3** embeddings, improving **search relevance by 21.7%** and **boosting query performance 25.6%** through unified re-ranking architecture
- Implemented a **Docker**-based data pipeline with **MongoDB** document storage on **AWS EC2**, reducing deployment setup time by 15.7% and deploying **CI/CD pipelines** with **Jenkins** to ensure high availability and automated testing, which resulted in an **18.7% reduction in deployment cycles** and supported a 2× increase in concurrent users from 5k to 10k.
- Implemented monitoring and observability with **Grafana** and **Prometheus**, providing real-time system insights and **reducing mean time to resolution (MTTR) by 25.3%**
- Streamlined development and quality assurance processes with **Postman** for API testing and **GitLab** for version control, resolving 82% of integration issues pre-deployment and improving team collaboration efficiency by 22%

***Machine Learning Engineer Intern* @ Data Mining & Security Lab, Beijing** Sep. 2022 - Jul. 2024
- Led research on **Federated Learning** and **Multi-View Multi-Label Machine Learning**, focusing on privacy-preserving feature fusion and multi-label classification. Published first-author paper in *IEEE Transactions on Big Data* 2025: "**Federated Multi-View Multi-Label Classification**" (DOI: 10.1109/TBDATA.2024.3522812)
- Devised solutions to complex data privacy challenges by proposing and developing the **FMVML** framework, a federated learning method enabling cross-view feature fusion and multi-label semantic classification, which outperformed all state-of-the-art methods, **improving Average Precision by 8.3% and lowering One Error by 14%**
- Utilized **Python**/**PyTorch** for model development and **Matlab** for signal processing; implemented data pipelines with **Pandas**/**Scikit-Learn**; produced publication-ready documents with LaTeX